

PRODUCT REQUIREMENTS DOCUMENT

DevFlow

The AI-Native End-to-End Project Management System

Version	1.0 — Initial Release
Status	Draft
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Owner	Viraj (Product & Engineering)
Audience	Engineering, Design, Investors

1. Executive Summary

DevFlow is an AI-native, end-to-end project management system purpose-built for small development teams (2–10 engineers). Unlike generic task trackers, DevFlow unifies project execution, real-time team communication, deep GitHub integration, and AI-powered workflow intelligence into a single coherent platform.

Where Linear is fast and Jira is powerful, DevFlow is both — and uniquely adds the AI layer that neither provides. DevFlow is not a tool you configure; it is a system that learns and adapts to how your team ships.

Vision

Every dev team ships faster when their tools think with them — not just for them.

1.1 The Problem

- ▶ Small teams juggle Jira/Linear for tasks, GitHub for code, Slack/Discord for chat, and separate dashboards for analytics — with zero coherence between them.
- ▶ Existing PM tools offer no real AI assistance in daily workflow — only cosmetic suggestions.
- ▶ GitHub activity (PRs, commits, reviews) is siloed from project progress, creating blind spots.
- ▶ Analytics in tools like Jira are rigid and uninformative for lean teams who need actionable signals.

1.2 The Solution

DevFlow collapses the fragmented dev-team stack into one platform:

- ▶ Kanban-first task management with multi-view support
- ▶ Built-in real-time team chat, native to the project workspace
- ▶ Deep GitHub sync: PRs, commits, and code reviews linked to tasks
- ▶ 6 AI-powered features embedded into the core workflow
- ▶ Rich, customizable analytics with workload, risk, and performance signals
- ▶ Notification bridge to Telegram and Discord (v1), Slack and Figma (v2)

2. Product Overview

2.1 Product Identity

Product Name	DevFlow
Tagline	Ship smarter. Together.
Category	AI-Native Project Management System
Primary Persona	Small engineering teams (2–10 devs)
Positioning	Faster than Linear. Smarter than Jira. Unified like nothing else.
Platform	Web application (desktop-first, mobile responsive)
Launch Target	Q4 2026 (v1 Public Beta)

2.2 Target Audience

Primary: Small Dev Teams (2–10 Engineers)

- Startups and indie product studios shipping software products
- Teams currently using a fragmented stack of Linear/Notion + Slack + GitHub
- Engineering leads who want visibility without micromanagement overhead
- Teams frustrated by Jira's complexity and Linear's limited AI capabilities

Secondary (Post-v1)

- Mid-size engineering orgs looking to standardize tooling across squads
- Dev-focused agencies managing client projects alongside internal repos

2.3 Core Value Propositions

Value Prop	Description	Key Differentiator
Unified Workspace	Tasks, chat, code activity, and analytics live in one place — no context switching.	Replaces 3–4 separate tools
GitHub-Native	PRs, commits, and reviews automatically surface inside tasks and dashboards.	Not a webhook — a deep sync
AI That Actually Works	6 AI features woven into the workflow: standup reports, sprint planning, task gen, and more.	Not a chatbot sidebar
Team Intelligence	Workload heatmaps, risk scoring, and performance metrics built for small teams.	Actionable, not vanity metrics

Communication Built-In	Real-time chat embedded in the workspace — not a third-party tab.	Kills the Slack context switch
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3. Goals & Success Metrics

3.1 Business Goals

- ▶ Achieve 500 active teams within 6 months of public beta launch
- ▶ Reach \$50K MRR within 12 months post-launch on freemium-to-paid conversion
- ▶ Establish DevFlow as the default PM choice for dev teams bootstrapping or scaling to Series A
- ▶ Maintain NPS > 60 in the first year — reflecting genuine delight, not just utility

3.2 Product Success Metrics (OKRs)

Metric	Definition	Target
Team Activation Rate	% of teams that complete onboarding + create first sprint	> 70% within 7 days of signup
AI Feature Adoption	% of active teams using at least 2 AI features weekly	> 50% of teams by Month 3
GitHub Integration Rate	% of v1 users who connect a GitHub repo	> 65% within first week
Retention (D30)	Teams still active 30 days post-signup	> 60%
Freemium Conversion	Free-to-paid upgrade within 60 days	> 12%
Avg Tasks/Team/Week	Indicator of core workflow adoption	> 15 tasks per team

3.3 Anti-Goals (What DevFlow Is NOT)

- ▶ Not a general-purpose project manager for non-dev teams (marketing, HR, etc.)
- ▶ Not a code editor, code review tool, or Git replacement
- ▶ Not a CRM, customer support tool, or product roadmap tool (v1)
- ▶ Not an enterprise-first platform — complexity and compliance features are out of scope for v1

4. Feature Specifications

4.1 Core Module: Project & Task Management

The primary interaction layer of DevFlow. Kanban-first, with the ability to switch to list and timeline views.

4.1.1 Kanban Board (Primary View)

- ▶ Drag-and-drop task cards across configurable columns (To Do, In Progress, In Review, Done)
- ▶ Custom column creation and color labeling per project
- ▶ Task cards show: assignee avatar, priority badge, GitHub PR status, due date, AI risk flag
- ▶ Bulk-select and move tasks across columns
- ▶ Swimlane support by assignee or priority

4.1.2 Task Detail

- ▶ Title, description (rich text with markdown support), priority, assignee, due date, labels
- ▶ Linked GitHub PRs, commits, and branches — auto-pulled from GitHub sync
- ▶ Threaded comments with @mentions and emoji reactions
- ▶ Activity log: all changes, GitHub events, and AI actions timestamped
- ▶ Sub-tasks with individual completion tracking
- ▶ Time estimates vs actual time (for analytics)

4.1.3 Additional Views (v1)

- ▶ List View — tabular task list with sortable columns and inline editing
- ▶ Timeline View — Gantt-style view for sprint and deadline planning

4.2 Core Module: Built-In Team Chat

Real-time communication embedded in the DevFlow workspace — not a third-party integration.

- ▶ Project channels: each project has a dedicated channel auto-created on project setup
- ▶ General workspace channel for cross-team announcements
- ▶ Direct messages (DMs) between team members
- ▶ Thread replies on messages to prevent channel noise
- ▶ Task linking in chat: paste a task ID and it unfurls into a rich preview
- ▶ AI standup bot posts automated daily summaries in the project channel
- ▶ File attachments, image previews, code snippet formatting with syntax highlighting
- ▶ Notification preferences: mute channels, set quiet hours, @mention alerts

4.3 Core Module: GitHub Deep Integration

DevFlow treats GitHub as a first-class data source — not a webhook listener.

- ▶ OAuth-based GitHub App installation per organization
- ▶ Auto-link PRs to tasks via branch naming convention: devflow/TASK-ID-description
- ▶ PR status badge on task cards: Open, Draft, In Review, Merged, Closed
- ▶ Commit activity timeline on task detail page
- ▶ AI PR Summary — auto-generated summary of what each PR changes and why (see 4.4.3)
- ▶ Repository overview on project dashboard: recent commits, open PRs, contributor activity
- ▶ GitHub issue sync: import GitHub issues as DevFlow tasks with bi-directional status updates

4.4 AI-Powered Features

All 6 AI features are powered by Claude (Anthropic API) and embedded contextually — no separate AI sidebar.

4.4.1 AI Task Generation from PRD

- ▶ Paste a PRD, feature description, or user story into a text field
- ▶ DevFlow AI generates a structured list of tasks, sub-tasks, and effort estimates
- ▶ User can review, edit, and bulk-import generated tasks into the active sprint
 - Input: plain text or markdown description
 - Output: structured task list with titles, descriptions, priority, and estimated hours

4.4.2 Smart Sprint Planning & Workload Balancing

- ▶ Before a sprint starts, AI analyzes team capacity (estimated hours vs. calendar availability)
- ▶ Flags over-allocated members and suggests task redistribution
- ▶ Recommends which backlog tasks to pull into the sprint based on priority and velocity history
- ▶ Shows predicted sprint completion likelihood as a percentage

4.4.3 AI Code Review Summaries

- ▶ When a PR is linked to a task and marked In Review, AI generates a plain-English summary
- ▶ Summary covers: what changed, files affected, potential concerns, and suggested review focus areas
- ▶ Posted as a pinned comment in the task's comment thread
- ▶ Saves reviewers 10–15 minutes per PR of orientation time

4.4.4 Automated Standup Reports

- ▶ Daily at a configurable time (default: 9:00 AM team timezone), AI posts a standup summary
- ▶ Per-member: tasks completed yesterday, tasks planned today, blockers detected
- ▶ Posted to the project's chat channel — readable by the whole team
- ▶ Each member can edit or add to their AI-drafted standup before it posts
- ▶ Weekly rollup report sent every Friday with sprint progress overview

4.4.5 Predictive Deadline & Bottleneck Detection

- ▶ AI monitors task aging, PR review wait times, and sprint velocity in real time
- ▶ Issues risk alerts on the dashboard: 'Task TASK-42 has been In Review for 4 days — deadline at risk'
- ▶ Bottleneck detection: identifies which team member or stage is creating the most flow interruption
- ▶ Suggested actions accompany every alert (reassign, reduce scope, extend deadline)

4.4.6 Natural Language Search

- ▶ Search across all tasks, comments, PRs, and chat messages using plain English
- ▶ Example queries: 'tasks blocked by authentication', 'what did Priya work on last sprint'
- ▶ AI interprets intent and returns ranked, context-aware results
- ▶ Replaces rigid filter systems with conversational discovery

5. Analytics & Reporting

DevFlow analytics are built for small team leads — not enterprise dashboards. Every metric shown maps to a decision you can actually make.

5.1 Analytics Modules

5.1.1 Team Workload Distribution Heatmap

- ▶ Visual grid: team members (rows) vs. days in sprint (columns)
- ▶ Color intensity represents task load: green (light), yellow (moderate), red (overloaded)
- ▶ Hover to see specific tasks contributing to load on any day
- ▶ Actionable: click any overloaded cell to reassign tasks directly

5.1.2 Individual Developer Performance Metrics

- ▶ Tasks completed per sprint, average task resolution time, PR review turnaround
- ▶ Streak tracking: consecutive sprints with on-time delivery
- ▶ Designed as a personal productivity tool — not a surveillance dashboard
- ▶ Each developer sees their own metrics by default; team lead sees aggregate

5.1.3 Deadline Adherence & Risk Scoring

- ▶ Per-task and per-sprint risk score (Low / Medium / High / Critical)
- ▶ Factors: task age, estimated vs elapsed time, PR status, assignee workload
- ▶ Risk score updated every 6 hours based on real-time GitHub and task activity
- ▶ Dashboard widget shows current sprint health with drill-down

5.1.4 Custom Dashboards

- ▶ Drag-and-drop widget layout per project and per user
- ▶ Available widgets: Sprint Burndown, Open PRs, Risk Radar, Workload Heatmap, Recent Chat Activity, Task Velocity
- ▶ Save and share dashboard layouts across the team
- ▶ Export to PDF or PNG for async reporting

5.2 Reports

Report Type	Description
Daily Standup Report	Auto-generated by AI, posted to project chat at configurable time
Weekly Sprint Summary	Progress vs. goal, risk flags, AI insights — sent every Friday

Sprint Retrospective Report	Velocity, completion rate, blockers, team load — generated at sprint close
GitHub Activity Report	PR throughput, review times, commit frequency per member per week
Custom Report Builder	Filter by date range, member, label, or project — export as CSV or PDF (v2)

6. Integrations

6.1 V1 Launch Integrations

6.1.1 GitHub (Deep Integration)

Covered in Section 4.3. This is a core product feature, not a lightweight plugin.

6.1.2 Telegram

- Bot-based integration: teams add @DevFlowBot to their Telegram group
- Receives: task status change notifications, AI standup summaries, risk alerts, PR merge notifications
- Supports slash commands: /tasks, /sprint, /status TASK-ID
- Configuration: per-project, per-channel, with notification type filtering

6.1.3 Discord

- DevFlow Discord App installable to any server
- Dedicated DevFlow channel per project, configurable via /devflow setup
- Same notification set as Telegram: task updates, standup reports, risk alerts, PR events
- Slash commands: /tasks open, /sprint status, /task TASK-ID
- Rich embeds with task cards including priority, assignee, and status

6.2 Planned V2 Integrations

Integration	Description
Slack	Workspace app with full feature parity to Discord/Telegram, plus Slack Workflow Builder support
Figma	Link design files to tasks; show Figma frame previews in task detail; notify on design updates
Vercel / Railway	Link deployments to sprints; surface deploy status on task cards; trigger deploys from DevFlow
Jira Import	One-click migration of Jira projects, epics, and issues into DevFlow workspace

7. Monetization & Pricing

7.1 Model: Freemium with Team Subscriptions

DevFlow uses a freemium model to drive top-of-funnel adoption among small teams, with a clear value cliff that motivates upgrade at the point of team growth and AI feature engagement.

7.2 Pricing Tiers

Plan	Includes	Price
Free	Up to 3 members, 2 projects, 5 AI actions/month, GitHub integration, Telegram & Discord notifications	\$0 / forever
Pro	Up to 10 members, unlimited projects, unlimited AI features, all analytics, custom dashboards, priority support	\$12 / member / month
Team (Annual)	Everything in Pro + SSO, audit logs, advanced permissions, dedicated onboarding, SLA support	\$96 / member / year

7.3 Upgrade Triggers (Free → Pro)

- ▶ Team reaches 3-member cap — natural forcing function
- ▶ AI usage limit hit (5 actions/month is low enough to show value, low enough to feel the ceiling)
- ▶ Access to workload heatmap and risk scoring — locked on Free
- ▶ Custom dashboard builder — locked on Free
- ▶ 14-day Pro trial auto-activated on team signup — no credit card required

8. Technical Architecture Overview

This section provides high-level direction for engineering. Detailed system design docs will follow separately.

8.1 Recommended Stack

Layer	Technology
Frontend	React + Vite + TypeScript
State Management	Zustand + TanStack Query
UI System	Shadcn UI + Tailwind CSS
Backend	NestJS (Node.js) + REST/WebSocket APIs
Database	PostgreSQL (primary) + Redis (caching, real-time pub/sub)
AI Layer	Anthropic Claude API (claude-sonnet-4-20250514)
Real-Time Chat	WebSockets via NestJS Gateway + Redis Pub/Sub
GitHub Integration	GitHub App (OAuth + Webhooks + REST API)
File Storage	AWS S3 or Cloudflare R2
Auth	JWT + OAuth 2.0 (GitHub, Google)
Deployment	Docker + Railway or Render (v1); AWS ECS (scale)

8.2 Key Architectural Decisions

- ▶ WebSockets for real-time chat and live task updates — Redis Pub/Sub for horizontal scaling
- ▶ GitHub webhooks ingested via a dedicated event queue (Bull/BullMQ) to avoid blocking API responses
- ▶ AI calls are async with optimistic UI — never block user interactions waiting for Claude API
- ▶ All analytics queries run against read replicas to protect write DB performance
- ▶ Multi-tenant data isolation: workspace-scoped row-level policies in PostgreSQL

9. Product Roadmap

9.1 V1 — Public Beta (Q4 2026)

- Kanban board, list view, timeline view
- Task management: full CRUD, sub-tasks, labels, priorities, due dates
- Built-in team chat with project channels and DMs
- GitHub deep integration (PR sync, commit linking, branch conventions)
- All 6 AI features (task gen, sprint planning, PR summaries, standups, risk detection, NL search)
- Analytics: workload heatmap, individual metrics, deadline risk scoring, custom dashboards
- Telegram and Discord integrations
- Freemium pricing: Free + Pro plans live

9.2 V2 — Growth Release (Q2 2027)

- Slack and Figma integrations
- Vercel / Railway deployment linking
- Jira and Linear import tools
- Custom report builder with CSV/PDF export
- Sprint retrospective AI report automation
- Mobile apps (iOS + Android) — read/notify-first, then edit
- Team plan: SSO, audit logs, advanced permissions

9.3 V3 — Platform (2028+)

- DevFlow API + public integrations marketplace
- AI agents: autonomous task creation from GitHub issues, PR-to-sprint auto-filing
- Enterprise tier: multi-org, SAML, SOC 2 compliance
- Embedded roadmapping module (strategy → sprint traceability)

10. Competitive Analysis

Tool	Strengths	Gaps vs. DevFlow
Linear	Fast, beautiful, dev-native	No built-in chat, minimal AI, no GitHub-linked analytics
Jira	Powerful, enterprise-grade, deep workflows	Complex setup, no AI, poor UX for small teams, no native chat
Notion	Flexible, all-in-one docs + tasks	Not dev-native, weak GitHub support, no real analytics
Asana	Strong project structure, workflow automation	Not dev-native, generic PM tool, no GitHub depth
DevFlow ✓	AI-native, GitHub-deep, unified chat + analytics	Purpose-built for dev teams — the gap none of the above fills

10.1 Positioning Statement

Position	For small dev teams who ship software products, DevFlow is the only project management system that combines AI-native workflow intelligence, deep GitHub integration, built-in team communication, and actionable analytics — in one place. Unlike Linear, which is fast but shallow, DevFlow is fast AND complete.
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11. Risks & Mitigations

Risk	Likelihood	Mitigation
AI feature quality underwhelms users	High	Invest in prompt engineering, allow AI output editing, ship progressively
GitHub integration breaks on API changes	Medium	Use official GitHub App framework; monitor deprecation notices; e2e tests on webhook flows
Real-time chat performance at scale	Medium	Redis Pub/Sub architecture designed for scale; load test pre-launch
Freemium users never convert to paid	High	Design clear value cliff; instrument AI usage limits to feel real; push 14-day trial aggressively
Small team loses focus, ships too much v1	High	Strict v1 scope lock; weekly scope review; defer v2 features ruthlessly
Competing tools copy AI features	Low (near-term)	Move fast; community + brand moat; deepen GitHub sync as defensible differentiator

12. Open Questions & Decisions Needed

- ▶ Should the AI standup report be opt-in per team, or opt-out by default? (User trust vs. adoption rate tradeoff)
- ▶ Does DevFlow support multiple GitHub repos per project in v1, or one repo per project?
- ▶ What is the data retention policy for chat messages on the Free tier?
- ▶ Will the Free plan support the Telegram/Discord integration, or is it a paid feature?
- ▶ Should individual developer performance metrics be visible to the team lead by default, or require explicit opt-in from each member?
- ▶ Localization / multi-language support — needed for v1 or v2?

Last updated: May 2026 — Viraj, DevFlow Product & Engineering